DAY 01 | PRE-READ

Talking to Computers

A gentle, no-code introduction to read before Day 1. About 15 minutes.

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Why read this?

Tomorrow you will write your first lines of Python. This short note gives you the big ideas in everyday language, so the class feels familiar instead of new. There is no code to memorise here. Just read, relax and think about the questions at the end.

1. What is a program?

A program is a list of instructions that a computer follows exactly, in order. Think of a recipe. A good recipe tells you what ingredients you need, the steps to follow, and what to do if something changes ("if the batter is too thick, add milk"). A computer is like a very fast cook who never gets tired but also never guesses. It does exactly what you write, nothing more and nothing less.

Key idea: computers are not smart, they are obedient. Clear instructions give clear results.

2. Why Python?

There are many programming languages, just as there are many spoken languages. Python is popular because it reads a lot like plain English. It is used to build websites, analyse data, automate boring tasks and build AI tools. Because it is friendly to beginners and powerful for experts, it is a great first language.

3. Five words you will hear tomorrow

Word	Everyday meaning
Code	The written instructions you give the computer.
Output	What the computer shows you back, like an answer on a calculator.
Variable	A labelled jar where you keep a piece of information, such as a name or a price.
Condition	A yes or no question that decides what happens next.
Loop	Doing the same step again and again until the job is done.

4. Ideas you already use every day

Labelled jars (variables)

In a kitchen you might have jars labelled "Sugar" and "Salt". The label stays the same, but what is inside can change. In programming, a variable works the same way: the label is the name, and the contents are the value.

Making decisions (conditions)

Every morning you think: "If it is raining, I will take an umbrella. Otherwise, I will take sunglasses." Programs make decisions in exactly this way.

Repeating steps (loops)

When you climb stairs, you repeat one action: lift foot, step up, until you reach the top. A loop lets a program repeat a step many times without writing it again and again.

Changing the form of something (type casting)

The text "25" written on a form and the number 25 look the same to us, but a computer treats them differently. You cannot add text the way you add numbers. Sometimes we must tell the computer, "treat this as a number".

5. Getting ready: a simple checklist

- Make sure you can sign in to a Google account. We will use **Google Colab**, which runs Python in your web browser.
- Open **colab.research.google.com** once to check it loads.
- Optional: install **VS Code** (code.visualstudio.com) and **Python** (python.org) on your laptop. Your trainer can help in class.
- Bring a notebook and pen. Writing small notes by hand helps memory.

6. Think about these before class

- Write down the steps for making a cup of tea. How many "if" decisions did you include?
- Name three things in your daily life you repeat until something changes.
- Why might a computer need to know whether "10" is text or a number?

Mindset tip: every programmer sees error messages every day. An error is not failure, it is the computer telling you exactly where to look. We follow the spirit of continuous small improvement.

Kaizen (改善): continuous improvement, one small step at a time